Understanding excess^3:

A Cross-Disciplinary Guide to Order-3 Synergistic Surplus

How to read, apply, and not over-claim a high-order dependence measure—without requiring a graduate course in information theory

Johel Padilla-Villanueva
Department of Environmental Health, University of Puerto Rico
ORCID: 0000-0002-5797-6931
`johelpadilla@gmail.com`
`joel.padilla2@upr.edu`

Guide v1.0 | July 2026
Repository: github.com/johelpadilla/excess3
DOI: 10.5281/zenodo.21385937

Abstract

Many scientific fields—ecology, epidemiology, physiology, climate science, neuroscience, and the social sciences—need tools that detect when several variables reorganise *together* in a way that pairwise correlations cannot fully explain. That idea is often called *high-order* or *synergistic* dependence. It is scientifically important and easy to misread.

This guide introduces **excess^3** (code name `excess3`): a pre-specified continuous proxy for order-3 synergistic surplus used in the Systemic Tau / RECD research programme. It is written for working scientists who are comfortable with statistics at the level of mutual information or regression, but who may not specialise in partial information decomposition (PID) or symbolic dynamics.

We prioritise intuition, analogies, and step-by-step calculation over brevity. Formal proofs, full synthetic batteries, and software edge cases are left to the methods paper [5], which remains the canonical specification. A dense Spanish introduction with the same validation evidence is available for Spanish-language readers [7]. Foundational Systemic Tau / RECD theory is cited only when needed for orientation [6]. The goal is not a thinner methods paper: it is a high-quality pedagogical route into a measure that is easy to misuse and powerful when used carefully.

Keywords: excess^3, synergistic information, high-order dependence, pedagogy, complex systems, multivariate statistics, RECD, ordinal patterns

Key idea

If you remember only one sentence: **excess^3 asks whether three (or more) variables carry joint structure that is not explained by looking at them two at a time**, and it answers with a continuous score plus a careful null test—not with a causal story and not with a full PID lattice.

Contents

1 Who this guide is for

1.1 Intended readers

This document is for scientists who may say things like:

- "I understand correlation and maybe mutual information, but PID papers feel opaque."
- "My system has three sensors / species / biomarkers, and something seems jointly wrong before each variable looks extreme."
- "I need a honest measure of high-order dependence, not another black-box AI score."

It is *not* a substitute for the methods paper when you implement the statistic, write a results section with surrogate p-values, or defend estimator choices under review. The family divides labour as follows:

Table 1: How the three excess^3 documents divide labour.

	This guide (EN)	**Methods (EN)**	**Intro (ES)**
Primary goal	Understanding and honest use	Canonical definition, nulls, synthetic validation	Dense motivation, reading, and evidence in Spanish
Tone	Pedagogical, cross-domain	Technical, pre-specified, reproducible	Expository prose with figures
Math depth	Progressive; every formula motivated	Compact and complete	Progressive with full numerical walk-through
Validation	Intuition + selected results	Full G0–G3 battery (claim-level)	Same battery integrated for interpretation
When to open	First contact; teaching; lab group	Implementation; manuscript methods; claims checklist	Spanish readers; courses; narrative results language
Cite for	Pedagogy only	Equations, nulls, checklist, software contract	ES exposition; still cite methods for the statistic

Learning outcomes. After this guide you should be able to (i) state the scientific question excess^3 answers in one sentence; (ii) compute Syn and Surp on a toy window and form $0.6\,\mathrm{Syn} + 0.4\,\mathrm{Surp}$; (iii) interpret a negative Δ without claiming "integration failed"; (iv) list what you must pre-specify before looking at p-values. You should *not* claim causation, a full PID atom, or Boolean $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$ from the reference construction.

1.2 How to read this guide

1. **If you only read three sections:** core idea (Section 2), six design decisions (Section 15), and FAQ (Section 16).
2. Read Sections 2–4 for the conceptual core (almost no formalism).
3. Read Sections 5–7 if you want the measure to stop feeling magical.
4. Read Sections 9–11 before applying it to real data.

5. Use Section 13 as a menu of domain framings.
6. Keep Section 15, Section 16, and the glossary (Section 21) open while reading papers or code.

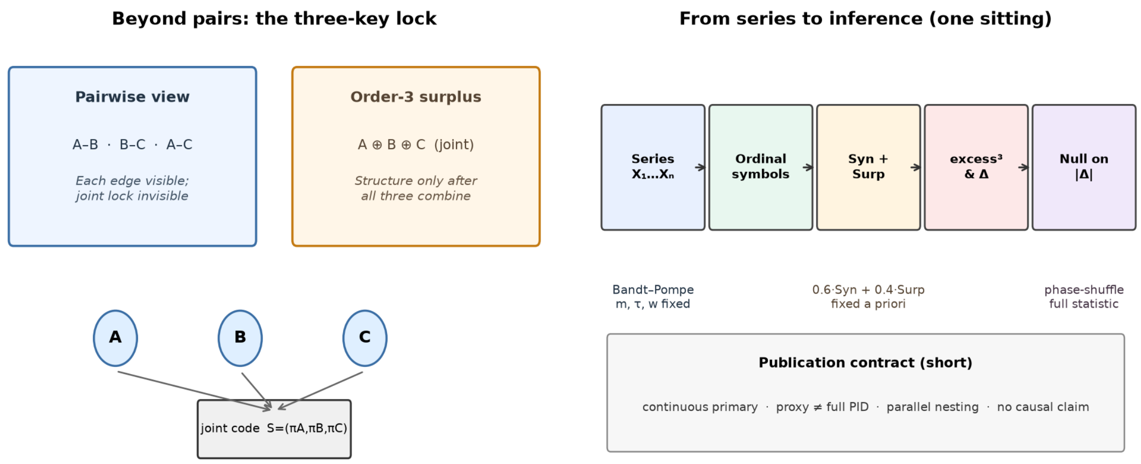

Figure 1: Pedagogical overview for first contact. **Left:** pairwise edges can look fine while a joint "three-key" code is invisible until all channels are read together. **Right:** pipeline from series to inference under the fixed contract (ordinal encoding, hybrid score, null on the full $|\Delta|$). This figure is teaching material; equations and the G0–G3 battery live in the methods paper [5].

1.3 What we deliberately do not do here

- We do **not** re-derive the full RECD / Systemic Tau framework.
- We do **not** dump every synthetic figure and hyperparameter table from the methods paper.
- We do **not** claim that excess^3 is "emergence proved" or "causation found".
- We do **not** lower the scientific standard of the methods paper; we raise the accessibility of the *ideas* behind it.

2 The core idea in one sitting

2.1 A three-variable lock

Imagine three binary indicators from a monitoring system:

$$A = \text{rainfall high?} \qquad B = \text{mosquito density high?} \qquad C = \text{fever cases high?}$$

You might find:

- A alone is only weakly related to C;
- B alone is only weakly related to C;
- A and B together tightly constrain C.

In everyday language: *the combination carries something the pairs do not.* In information language: there is **order-3 synergistic structure** (or synergistic surplus).

A related but different pattern is the opposite: A, B, and C all rise and fall together because of one shared driver (temperature, seasonality, a common shock). Then pairs already explain most of the joint behaviour. That is more like **redundancy** or common-cause locking than pure order-3 synergy.

Key idea

High-order dependence is about **organisation among variables**, not about whether any single variable has large variance. A quiet system can still reorganise its joint code; a noisy system can look "active" without true multi-variable structure.

2.2 What excess^3 tries to answer

excess^3 is an operational answer to a narrow question:

> *In a sliding window of multivariate observations (after a symbolic encoding), how much joint surplus remains after we account for pairwise structure and for simple independence?*

It returns a non-negative continuous score over time. Larger values mean more of that surplus under the chosen encoding and window. Contrasts Δexcess3 (before vs after an event; condition A vs B) are usually more scientifically useful than absolute levels.

2.3 The one-line formula (preview)

You will understand every term by Section 7. For orientation only:

$$\text{excess3} = 0.6\,\text{Syn} + 0.4\,\text{Surp}. \tag{1}$$

- Syn = residual multi-information after subtracting a pairwise baseline ("what pairs do not explain").
- Surp = how much more often joint symbol combinations occur than under full independence.
- The weights 0.6 and 0.4 are **fixed in advance**. They are not optimised on your scientific dataset.

Caution

If you re-tune the weights on the same data you later claim results from, you have turned a pre-specified test statistic into a flexible descriptor. That may still be exploratory—but it is no longer the same methodological object.

3 Analogies that carry the mathematics

Formal definitions come later. First, three analogies that map cleanly onto what the measure does.

3.1 The safe with two keys vs three keys

- **Pairwise world:** any two keys open something useful. Looking at pairs is enough.
- **Order-3 synergistic world:** the vault opens only when three keys are combined in the right pattern. Any pair looks uninformative; the triple is decisive.
- **Common-driver world:** one master key turns all locks. Pairs look extremely informative; the "extra" from the triple is small once pairs are accounted for.

excess^3 is closer to detecting the *three-key* situation, while still remaining sensitive to some joint surprises that pure pair-subtraction might underplay (that is why it has two ingredients).

3.2 Orchestra vs soloists

If each instrument plays louder, univariate early-warning tools may fire. If the instruments start coordinating in a new ensemble pattern—without any soloist becoming extreme—you need a relational measure. excess^3 is one way to listen for *ensemble code* among three or more parts.

3.3 Password vs checksum

A password that is the XOR of two other bits is a classic synergy toy: each bit alone is noise; together they determine the third. A checksum that merely copies a shared flag is redundancy. Section 8 calculates the password case by hand.

In your field

Ecology. Species A and B may each correlate weakly with a collapse indicator C, yet the combination "A low and B high" may lock C.
Physiology. Heart-rate, respiration, and blood-pressure patterns may reorganise jointly before any channel crosses a clinical threshold.
Climate. Three regional indices can reorganise their co-occurrence code near a regime shift even if pairwise correlations change only mildly.

4 Why pairs are not enough

4.1 What pairwise tools already give you

Scientists already use excellent pairwise tools:

- Pearson / Spearman correlations;
- mutual information $I(X;Y)$;
- transfer entropy and Granger-style directed tests;
- network edges based on significant pairs.

These answer: *how are two variables related?* They do not fully answer: *is there structure that only appears when three or more are considered jointly?*

4.2 A minimal counterexample (XOR intuition)

Suppose three binary variables satisfy $C = A \oplus B$ (XOR), with A and B fair coins. Then:

- $I(A;C) = 0$, $I(B;C) = 0$, $I(A;B) = 0$ (pairs look independent);
- but (A, B) determines C completely.

Any analysis that only draws a network of pairwise edges will conclude "nothing is related." A high-order lens sees a tight joint law.

Key idea

Pairwise null results do **not** imply "no dependence in the system." They imply "no dependence visible at order 2 under this estimator."

4.3 The opposite trap: calling every triple "synergy"

Not every interesting three-variable pattern is synergistic surplus. Shared seasonality, one latent shock, or a strong common driver can make all pairs light up. That is scientifically important—but it is a different geometry. excess^3 tries to emphasise residual joint structure after a pairwise baseline, plus independence surprise. It is still a **proxy**, not a court verdict that "true synergy exists in nature."

5 A gentle information primer

This section is intentionally elementary. Skip it if entropy and mutual information are already fluent for you; otherwise it is the bridge into Section 7.

5.1 Uncertainty as a number: entropy

Definition (plain language)

Entropy $H(X)$ measures how unpredictable a variable is under a chosen unit (here: bits). A fair coin has $H = 1$ bit. A coin that always lands heads has $H = 0$.

For a discrete variable with probabilities $p(x)$,

$$H(X) = -\sum_{x} p(x) \log_2 p(x). \tag{2}$$

You do not need deep measure theory to use this: plug in frequencies from a window of symbols.

5.2 Shared information: mutual information

Definition (plain language)

Mutual information $I(X;Y)$ measures how much knowing X reduces uncertainty about Y (and vice versa). It is zero when X and Y are independent under the estimated distribution.

$$I(X;Y) = H(X) + H(Y) - H(X,Y). \tag{3}$$

Interpretation checklist:

- $I = 0$: no detectable pairwise dependence (under this estimator);
- larger I: stronger pairwise dependence;
- I is symmetric and non-negative (for the usual Shannon version).

5.3 More than two variables: total correlation

Definition (plain language)

Total correlation (multiinformation)

$$\mathrm{TC}(X_1, \ldots, X_N) = \sum_{i=1}^{N} H(X_i) - H(X_1, \ldots, X_N)$$

measures *total* shared information among N variables. It is zero only under full mutual independence.

TC is large when the joint is far more organised than the product of marginals. Crucially, TC does **not** by itself tell you whether that organisation is mostly pairs or mostly higher-order structure.

5.4 The missing piece: a pairwise baseline

To isolate something more like "beyond pairs," subtract a baseline built from pairwise mutual informations. In the excess^3 construction the baseline is scaled mean pairwise MI:

$$I_{\text{pair}} = (N-1) \cdot \overline{I}_{ij}, \tag{4}$$

where $\overline{I}_{ij}$ is the average of $I(X_i; X_j)$ over pairs. The factor $(N-1)$ puts the baseline on a scale comparable to TC [6, 5].

Then

$$\mathrm{Syn} = \left[\mathrm{TC} - I_{\text{pair}}\right]_+ = \max\{\mathrm{TC} - I_{\text{pair}}, 0\}. \tag{5}$$

If pairs already explain the multiinformation, Syn collapses toward zero. If multiinformation outruns the pairwise baseline, Syn rises.

Caution

Syn is a **residual construction**, not a mystical proof of emergence. Different baselines yield different residuals. The methods paper fixes one baseline aligned with the reference software.

5.5 Independence surprise

Even after residual multiinformation, it helps to ask a second question:

> *Do particular joint combinations occur more often than the product of marginals would allow?*

That is Surp: a weighted sum of positive log-ratio surprises for joint symbol tuples. It is complementary to Syn:

- Syn lives in multiinformation-minus-pairs geometry;
- Surp lives in "joint co-occurrence vs independence" geometry.

6 From raw series to symbols (why ordinal encoding?)

6.1 The practical problem

Real variables have different units, drifts, and heavy tails. Direct plug-in entropies on raw continuous values need binning choices that can dominate the result.

6.2 Bandt–Pompe ordinal patterns in one paragraph

Bandt–Pompe encoding replaces a local segment of a series by the **rank pattern** of its values [1]. For embedding dimension $m = 3$, each window of three samples becomes one of $3! = 6$ possible order patterns ("up-up", "peak", "trough", etc.).

Why scientists like this:

- invariant to monotone transformations of each channel;
- focuses on relational shape rather than absolute amplitude;
- yields a discrete alphabet where information quantities are straightforward to estimate in short windows.

Key idea

excess^3 as used in RECD is typically computed on **ordinal symbol streams**, not on raw millimetres of mercury or raw case counts. That is a feature for cross-domain comparison, and a limitation if your scientific claim is about amplitudes rather than order patterns.

6.3 Windows

Everything is estimated inside a sliding window of length w (reference software often uses $w = 13$ with $m = 3$). Think of excess$^3(t)$ as a local "joint organisation score" ending at time t, not as a single number for the whole study.

7 Building excess^3 brick by brick

7.1 The pipeline

1. Start with N aligned time series (the interesting scientific case is $N = 3$, but the formulas extend).
2. Encode each series into ordinal symbols.
3. In a window, estimate entropies of each channel, all pairs, and the full joint.

4. Compute TC, I_{pair}, Syn, Surp.
5. Combine: $\text{excess3} = 0.6\,\text{Syn} + 0.4\,\text{Surp}$.
6. Optionally threshold: $\Phi_3 = \mathbf{1}\{\text{excess3} > \theta_3\}$.

7.2 Component table

Table 2: The moving parts of excess^3 in plain language.

Symbol	Plain meaning	If it is high...
TC	Total shared information among the N symbol streams	The joint is organised far beyond independence
I_{pair}	Pairwise baseline (scaled mean pairwise MI)	Pairs already carry a lot of structure
Syn	$\max(\text{TC} - I_{\text{pair}}, 0)$	Multiinformation exceeds what pairs explain
Surp	Positive surprise of joints vs product of marginals	Specific joint combinations are over-represented
excess^3	$0.6\,\text{Syn} + 0.4\,\text{Surp}$	Hybrid synergistic surplus (continuous)
Φ_3	Binary tick if $\text{excess}^3 > \theta_3$	Only a discretisation for counts/displays

7.3 Why a hybrid?

A single pure atom from a full PID lattice is conceptually cleaner—and often statistically harder at the sample sizes of ecology, epidemiology, or field physiology. The hybrid is a deliberate engineering choice:

- Syn pushes against "pairs already explained it";
- Surp keeps sensitivity to joint co-occurrence structure;
- fixed weights keep the object pre-specified and comparable across studies.

This is why the methods paper calls excess^3 a **computable proxy** rather than "the" synergy atom [5].

8 Worked example: the XOR-like window

This is the pedagogical heart of the guide. Numbers are tiny on purpose.

Worked example

Setup. Window length $w = 8$, three symbol streams with alphabet $\{0, 1\}$. The eight rows implement a password-like lock: each pair is uninformative, but the triple is tightly constrained (XOR geometry).

$\pi^{(1)}$	$\pi^{(2)}$	$\pi^{(3)}$	
0	0	0	
0	1	1	
1	0	1	
1	1	0	each of these four joint types appears twice
0	0	0	
0	1	1	
1	0	1	
1	1	0	

Step 1 — single-channel entropies. Each channel is half 0s and half 1s, so $H(\pi^{(i)}) = 1$ bit.

Step 2 — joint entropy. There are four joint types, each with probability 1/4, so the joint entropy is $H(\pi^{(1:3)}) = 2$ bits.

Step 3 — total correlation.

$$\mathrm{TC} = (1 + 1 + 1) - 2 = 1.$$

Step 4 — pairwise baseline. Every pair is balanced in a way that pairwise MI vanishes, so $\bar{I}_{ij} = 0$ and $I_{\mathrm{pair}} = 0$.

Step 5 — Syn.

$$\mathrm{Syn} = [1 - 0]_+ = 1.$$

Step 6 — Surp. Under independence, each joint triple would have probability $(1/2)^3 = 1/8$. Empirically each observed type has probability 1/4. The log-ratio is

$$\log_2 \frac{1/4}{1/8} = 1,$$

and four types each contribute $(1/4) \cdot 1$, so $\mathrm{Surp} = 1$.

Step 7 — excess^3.

$$\mathrm{excess3} = 0.6 \cdot 1 + 0.4 \cdot 1 = 1.0.$$

Key idea

In the XOR window, **both** ingredients fire. Pairs see nothing; the joint sees everything. That is the pedagogical prototype of order-3 synergistic surplus.

8.1 Contrast: pure common driver

If all three channels are identical (only joints like $(0, 0, 0)$ and $(1, 1, 1)$ appear), pairs become extremely informative. Then I_{pair} can absorb most of TC, so $\mathrm{Syn} \to 0$, while Surp may remain large because the joint is still far from independence. excess^3 then lives mainly on the surprise leg. This is intentional: redundant locking and XOR-like locking are different geometries [5].

9 How to read magnitudes and signs

9.1 Absolute level vs contrast

Table 3: Reading excess^3 values without overclaiming.

Pattern	**Honest reading**
excess$^3(t) \approx 0$	Joint structure is largely explainable by pairs or independence under this encoding and window.
Sustained elevation (often $\gtrsim 0.10$–0.20, domain-dependent)	Suggests detectable synergistic surplus—not a universal clinical cut-off.
Δexcess3 pre/post	Usually the scientific object of interest (did organisation change?).
Negative Δ after a reorganisation	Possible and informative: stronger common coupling can inflate pairwise baselines and shrink residual Syn.
Binary Φ_3 alone	Incomplete if continuous curves are ignored; thresholds can saturate.

Caution

There is **no universal threshold** that means "synergy happened" across ecology, cardiology, and climate. Always interpret with a null (Section 11) and report encoding parameters (m, τ, w).

9.2 Why a change can be negative even when "something real happened"

This is one of the most important practical lessons from synthetic validation [5]. In a shared-latent generator, when common coupling strengthens after an event, pairwise mutual informations rise. Because Syn subtracts a pairwise baseline, residual multiinformation can *fall* even though the system truly reorganised.

Therefore inference is performed on $|\Delta\text{excess3}|$ (two-sided extremity under surrogates), not on a forced positive sign.

Key idea

A negative Δexcess3 is not automatically a failed experiment. It can mean: **pairwise structure absorbed the change**. Report the sign, explain it, and test extremity against a null.

9.3 Continuous primacy

The binary indicator

$$\Phi_3(t) = \mathbf{1}\{\text{excess3}(t) > \theta_3\} \tag{6}$$

is useful for tick counts and raster-like displays. It is a **secondary** discretisation. Continuous excess^3 and Δexcess3 are primary for scientific contrasts. In synthetic batteries, a low reference θ_3 can saturate occupancy while continuous Δ still discriminates arms [5].

10 What excess^3 is not

This section exists to protect your reputation in interdisciplinary review.

1. **Not causation.** Co-occurrence beyond pairs is not a directed mechanism. Causal claims need design, lags, interventions, or explicit causal models.
2. **Not a full PID synergy atom.** PID isolates unique, redundant, and synergistic atoms on a lattice [14, 2]. excess^3 is a practical proxy in that neighbourhood, not a replacement theorem.
3. **Not proof of strong emergence.** "Irreducible in this statistical sense" is weaker than metaphysical emergence. Keep the language operational.
4. **Not a free lunch for tiny samples.** Joint symbol distributions need enough window mass. Short T, large N, or large alphabets inflate variance and bias.
5. **Not interchangeable with O-information without care.** O-/S-information provide global redundancy-vs-synergy decompositions [9]. They are relatives, not synonyms.
6. **Not "whatever the weights say after tuning."** Pre-specification is part of the object. If you retune, say so and treat claims as exploratory.

Table 4: Positioning for non-specialists: what family of questions each tool answers.

Measure	**Main question**	**Cost**	**Relation to excess^3**
Interaction information	Signed three-variable interaction (can be negative)	Low	Conceptual cousin; not identical
O-information	Global redundancy vs synergy balance	Medium	More global; not locked to order exactly 3
PID synergy atom	Pure synergistic atom on the lattice	High	Conceptual target; often hard to estimate
excess^3	Pre-specified hybrid order-3 surplus proxy	Low–med.	This guide's object

11 How inference works (without the full methods appendix)

11.1 The question a null must answer

You observed Δexcess3 between two regimes. Could a process with similar single-channel spectra but destroyed cross-variable phase relationships produce a Δ at least as extreme?

11.2 Phase-shuffle in plain language

A practical surrogate:

1. Fourier-transform each channel;
2. randomise phases (preserving power spectrum, roughly);

3. invert, re-encode ordinal symbols, recompute excess^3 and Δ;
4. repeat B times to build a null distribution;
5. place the observed $|\Delta|$ in that distribution.

This breaks fine-grained cross-channel organisation while retaining much of each channel's autocorrelation structure—a better straw man than pure white noise for many time-series settings [11, 5].

Caution

Do not only compare "mean of observed windows vs mean of null means" without ranking the **full statistic** you care about. The methods paper emphasises testing Δexcess3 (or the chosen contrast) against the surrogate distribution of that same contrast.

11.3 What synthetic validation taught us (selected)

Under a controlled battery (near-null independent AR channels; planted shared-latent reorganisation; coupled logistics; pure pairwise control), the methods paper finds, in short:

- near-null arms stay near zero Δ with non-extreme surrogate p-values;
- planted joint reorganisation yields extreme $|\Delta|$ at high rates;
- the sign under shared-latent strengthening can be negative;
- continuous Δ discriminates even when binary occupancy saturates at a low threshold.

Those results justify careful use; they do **not** license universal thresholds on real observational data. See [5] for numbers, figures, and claim gates.

12 A minimal path into Systemic Tau / RECD

You can use excess^3 as a standalone high-order feature. In the broader programme it is Level 3 of a nested ordinal hierarchy [6]:

- **Systemic Tau** (τ_s): a continuous "thermometer" of multivariate rank concordance (relational coherence).
- **RECD**: a discrete intrinsic clock advanced by ordinal conjunctions at several depths.
- **Level 1 (Φ_1):** coincidence-like events.
- **Level 2 (Φ_2):** persistent pairwise relational events.
- **Level 3:** synergistic surplus—continuous excess^3 and optional binary Φ_3.

Operative nesting (important). It is tempting to imagine a logical tower $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$. In the reference construction the levels are computed **in parallel**. Order-3 irreducibility is statistical (via the excess construction), not a hard Boolean implication. This is design decision 6 in Section 15.

Key idea

If you only need a high-order dependence feature for a classifier or an early-warning panel, you can start with excess^3 alone. If you need the full Kairos-style clock narrative, read the framework paper after this guide—not before.

13 Field vignettes: how to talk about it in your domain

These vignettes are **framing aids**, not empirical claims about particular datasets.

13.1 Ecology and conservation

In your field

Question. Do three indicators of ecosystem state reorganise their joint ordinal code before a regime shift, even if no single indicator has crossed a classical early-warning threshold?

Pair with classical variance / lag-1 autocorrelation tools [10, 3]. Report excess^3 as a relational complement, not a replacement. Watch for seasonality as a common driver that can inflate pairs.
Allowed: "joint ordinal reorganisation beyond pairs." *Forbidden:* "proved ecosystem emergence / tipping caused by synergy."

13.2 Epidemiology and environmental health

In your field

Question. Do climate, vector, and incidence symbols develop beyond-pairwise co-occurrence structure around outbreak onsets?

Be explicit: excess^3 does not identify which variable "causes" cases. It flags joint reorganisation worth mechanistic follow-up.
Allowed: "beyond-pairwise co-occurrence around onset." *Forbidden:* "climate caused incidence via excess^3."

13.3 Physiology and critical care

In your field

Question. Do cardiorespiratory ordinal patterns show synergistic surplus changes around decompensation windows?

Continuous primacy matters: binary alarms may sit silent while continuous excess^3 drifts. Thresholds must be justified in-domain, never copied blindly from software defaults.
Allowed: "continuous surplus drift under a pre-specified null." *Forbidden:* "Φ_3 alarm proves decompensation."

13.4 Climate and Earth system science

In your field

Question. Near a potential tipping element, do three regional indices show residual joint structure beyond pairwise teleconnection maps?

Compare against phase-shuffle nulls that preserve much of each index's spectrum. Document encoding (m, τ, w) as carefully as a filter choice.
Allowed: residual joint structure vs pairwise maps. *Forbidden:* "tipping confirmed by synergy score alone."

13.5 Neuroscience and cognitive science

In your field

Question. Is there order-3 synergistic surplus among three neural or behavioural readouts during integration epochs?

Readers in this field often expect PID language. State clearly that excess^3 is a proxy neighbour of synergy atoms, not a lattice decomposition [14, 9].
Allowed: "proxy for order-3 surplus (not a PID atom)." *Forbidden:* "we measured the synergy atom of consciousness."

13.6 Social and economic systems

In your field

Question. Do three market or mobility series reorganise jointly around shocks in a way edge-networks miss?

Common shocks are the main confounder. Negative Δ after synchronisation spikes is a live possibility; interpret with the pairwise baseline story.
Allowed: "joint reorganisation; possible pairwise absorption." *Forbidden:* "markets lost integration because $\Delta < 0$."

14 A first-analysis workflow

1. **State the scientific contrast** before computing. Pre/post event? Treatment vs control? Seasonal baseline vs anomaly?
2. **Choose $N = 3$ channels** with a joint hypothesis (not three random leftover sensors).
3. **Fix encoding parameters a priori:** m, τ, w, stride. Write them down before looking at p-values.
4. **Compute continuous excess$^3(t)$** and the pre-specified contrast Δ.
5. **Run phase-shuffle (or design-based) nulls** on that same contrast; report median p and fraction extreme if using ensembles.

6. **Report Syn and Surp separately** at least in a supplement: which leg moved?
7. **Show continuous curves**; treat Φ_3 as secondary.
8. **Language audit:** dependence/organisation/surplus—not causation/emergence proved.
9. **Sensitivity:** at least one alternate window length or m if sample size allows.
10. **Point to the methods paper** for estimator details and software alignment.

14.1 Minimum reporting checklist (copy into lab notes)

- Series length T, number of channels N, sampling rate;
- Bandt–Pompe (m, τ), window w, stride;
- weights $(0.6, 0.4)$ and statement that they were not tuned on this dataset;
- primary statistic: continuous excess^3 / Δexcess3;
- θ_3 only if Φ_3 is used, with saturation checks;
- null model, B, two-sided vs one-sided rule;
- software commit / version when available;
- explicit non-claims (not causal, not full PID).

15 Six design decisions (the fixed contract)

When readers first meet excess^3, six questions almost always reappear. The short answers below are the **methodological contract** shared with the methods paper [5]: they are design choices, not optional footnotes. If you remember nothing else from this guide's middle chapters, remember these six.

1. Why 0.6/0.4**?** The weights $(\alpha_{\text{Syn}}, \alpha_{\text{Surp}}) = (0.6, 0.4)$ are **heuristic constants fixed a priori**. They lean slightly toward the residual multiinformation leg (Syn) over independence surprise (Surp), but they are *not* fitted by cross-validation, grid search, or any other optimiser on the scientific dataset under test. That pre-specification protects *falsifiability*: excess^3 stays a declared statistic rather than a dial you turn until a favourite p-value appears. You may pre-register a different pair *before* looking at the hypothesis data; silent re-weighting on the claim dataset is not compatible with presenting the canonical excess^3 object.

2. Syn versus Surp. Syn $= [\text{TC} - I_{\text{pair}}]_+$ asks: after we **subtract** a pairwise mutual-information baseline from total correlation, how much multiinformational organisation remains? Surp asks a different question: which joint symbol tuples are **more frequent than under complete independence** (p_{obs} vs. $\prod_i p_i$), counting only positive log-ratio contributions? The hybrid keeps both lenses visible. XOR-like locks tend to light both; pure common-driver locking often collapses Syn while Surp stays large (Section 8).

3. excess^3 versus Φ_3. Continuous excess3(t) is the **primary** Level-3 readout: trajectories, gradients, and contrasts Δexcess3 live on the continuous scale. Binary $\Phi_3 = \mathbf{1}\{\text{excess3} > \theta_3\}$ exists only for tick counts, raster-like displays, and rate summaries. If θ_3 sits below the bulk of the continuous distribution, Φ_3 occupancy can saturate even when Δ still separates regimes—that is a threshold issue, not proof that "Level 3 did nothing" (Section 9).

4. Not a complete PID. excess^3 is a **computable proxy** for order-3 synergistic surplus, not a full partial information decomposition. It does not isolate unique, redundant, and synergistic atoms of a PID lattice [14, 2]. The natural theoretical extension is to replace the hybrid formula by a stable PID synergy atom when sample size and estimator quality allow; until then, the hybrid is the operational contract (Section 10).

5. The null. Inference targets the **full declared contrast**—usually $|\Delta\text{excess3}|$ under a two-sided Monte Carlo rule—not a vague "mean vs mean of means." The default surrogate is independent phase-shuffle (or any pre-specified alternative that preserves relevant univariate structure while **breaking cross-channel dependence**). Ranking the observed contrast inside the surrogate cloud is the protocol; mean-versus-mean alone is an anti-pattern (Section 11).

6. Operative nesting. In RECD language one may speak of deeper levels as stronger claims. In software, Φ_1, Φ_2, and excess^3/Φ_3 are computed **in parallel** from the same symbolic window. There is no hard Boolean implication $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$ in the reference code. Order-3 irreducibility is inferred *statistically* through the excess construction (Syn after the pairwise baseline, plus Surp), not by forcing a chain of logical gates (Section 12).

Key idea

Weights fixed before the claim data; Syn subtracts pairs, Surp scores over-represented joints; continuous first, binary only for ticks; proxy not full PID; nulls on full Δ; RECD levels in parallel, not a Boolean ladder. That is the whole design contract in one breath.

16 Common misconceptions (FAQ)

Section 15 already settles the six design decisions. The questions below cover remaining confusions that still show up in seminars and reviews.

Q1. Is excess^3 just "the synergy from PID"? No—see also design decision 4. It is a hybrid proxy for stable pre-specified use when full PID estimation is fragile or unnecessary for the claim.

Q2. If excess^3 is high, did we find emergence? You found elevated residual joint organisation under a specific encoding. That can support an emergence-*related* discussion; it does not settle philosophical emergence.

Q3. Why not always use interaction information alone? Interaction information is a valuable cousin and can be negative [4, 12]. excess^3 is a different operational object: non-negative hybrid residual plus independence surprise, with fixed weights and a software-aligned pipeline.

Q4. My Δ is negative but the system clearly changed. Failed? Not necessarily. Check whether pairwise dependence exploded. Look at Syn vs Surp separately. Test $|\Delta|$ against surrogates.

Q5. Can I optimise the 0.6/0.4 weights with cross-validation? You can explore weight sensitivity in a methods supplement. If weights are chosen to maximise your scientific effect on the same data, label the analysis exploratory and do not present it as the pre-specified excess^3 statistic (design decision 1).

Q6. Do I need the full RECD story to publish with excess^3? No. You need honest definition, nulls, and domain interpretation. RECD context is optional framing (design decision 6: levels need not be a Boolean tower).

Q7. Is binary Φ_3 good enough for a paper? Usually not by itself. Continuous trajectories and contrasts should lead; binary ticks can accompany them (design decision 3).

Q8. What sample size do I need? There is no universal n. Joint alphabets grow fast. Prefer longer series, modest m, and surrogate tests. If the joint histogram is mostly empty, you are not estimating high-order structure—you are estimating sparsity.

Q9. Can excess^3 replace early-warning indicators? No. Classical EWS tools remain powerful for many univariate bifurcation routes [10]. excess^3 is a relational complement.

Q10. Where is the code? Reference implementation lives in the Systemic Tau / RECD software stack (`recd_levels` / `_synergy_and_surprise`); the methods paper and its toy script document alignment [5, 8].

17 A second worked sketch: reading a pre/post contrast

Suppose a laboratory records three ordinalised channels around a known perturbation at time $t^\star$. They compute:

$$\text{mean excess}^3\text{ before: }1.83 \qquad \text{mean after: }1.73 \qquad \Delta = -0.10.$$

Phase-shuffle nulls put $|\Delta|$ in the extreme tail (e.g. median $p \approx 0.01$ across realisations in a synthetic analogue).

Bad write-up. "Synergy decreased, therefore integration failed."

Better write-up. "The pre-specified contrast Δexcess3 was negative and surrogate-extreme. Separate inspection shows pairwise mutual information increased after $t^\star$, shrinking the residual Syn term. We therefore interpret the change as a reorganisation toward stronger common coupling rather than as 'nothing happened.' Continuous excess^3 remained the primary readout; binary occupancy was saturated at the reference threshold."

This narrative mirrors lessons from the shared-latent synthetic arm without requiring the reader to digest the full validation suite first [5].

18 Teaching notes (for courses and lab meetings)

18.1 A 45-minute reading-group plan

1. 10 min: XOR intuition on the whiteboard (no software).
2. 10 min: compute TC, Syn, Surp by hand from Section 8.
3. 10 min: discuss negative Δ and pairwise absorption.
4. 10 min: domain vignette chosen by the group (Section 13).
5. 5 min: language audit (Section 10).

18.2 A half-day workshop plan

1. Morning: this guide through Section 9.
2. Midday: run the public toy example script from the methods companion.
3. Afternoon: each participant frames one contrast in their data (without p-hacking weights).
4. Close: map open questions to the methods paper sections (nulls, estimation, limitations).

18.3 What to assign as required reading

- **Everyone:** this guide.
- **Implementers / methods reviewers:** methods paper [5].
- **Spanish-language groups:** dense introduction [7] (same contract and validation evidence).
- **Readers who need the full clock narrative:** framework paper [6].
- **Optional classical background:** Bandt–Pompe [1]; early-warning review [10]; PID entry points [14, 12].

19 Limitations (honest, still pedagogical)

- **Encoding dependence.** Ordinal patterns emphasise order, not amplitude. If your science is amplitude-first, say so and consider complementary measures.
- **Estimator dependence.** Plug-in entropies are transparent but biased in small samples. Changing estimators changes numeric scales.

- **Proxy gap.** Distance to a true PID synergy atom is not zero and is context-dependent.
- **Computational growth.** Moving from order 3 toward excess^k becomes expensive quickly.
- **Null mismatch.** Phase-shuffle is powerful, not universal. If your alternative hypothesis is a different kind of structure, design a matching null.
- **Interdisciplinary rhetoric risk.** Words like *synergy*, *emergence*, and *integration* travel poorly across fields. Prefer operational phrasing in results sections.

20 Where to go next

1. **Methods companion.** Canonical equations, null protocol, synthetic G0–G3 validation, reporting checklist for manuscripts [5].
2. **Spanish introduction.** Dense motivation and the full G0–G3 figure set in Spanish [7].
3. **Framework paper.** Systemic Tau as relational thermometer; RECD as intrinsic discrete time; how Level 3 sits in the paradigm [6].
4. **Software.** Reference implementation and reproducibility notes [8] (`https://doi.org/10.5281/zenodo.20576241`).
5. **Domain papers.** Apply the workflow of Section 14 to your system; do not expect this guide to contain your field's empirical result.

Key idea

The highest-quality use of excess^3 is boring in the best sense: pre-specified, continuous-first, null-tested, modest in language, and clear about being a proxy for high-order dependence rather than a metaphysical detector.

21 Glossary

Alphabet

The set of possible symbols after encoding (for Bandt–Pompe with embedding m, size $m!$).

Bandt–Pompe / ordinal pattern

Local rank-order encoding of a real-valued series [1].

excess^3 / `excess3`

Hybrid continuous proxy $0.6\,\mathrm{Syn} + 0.4\,\mathrm{Surp}$ for order-3 synergistic surplus.

High-order dependence

Statistical organisation involving three or more variables that is not fully visible from pairs alone.

I_{pair}

Pairwise baseline: scaled mean pairwise mutual information.

Mutual information
: Shared information between two variables; zero under independence.

O-information / S-information
: Global decompositions emphasising redundancy-like vs synergy-like modes [9].

Phase-shuffle surrogate
: Null series that roughly preserve spectra while destroying cross-phase organisation.

PID
: Partial information decomposition: lattice of unique, redundant, and synergistic atoms [14].

Φ_3
: Binary tick $\mathbf{1}\{\text{excess3} > \theta_3\}$; secondary to continuous excess^3.

Pre-specification
: Fixing formula and weights before examining the scientific dataset's target claims.

RECD
: Discrete Extramental Clock: hierarchical ordinal event time in the paradigm [6].

Redundancy (informal)
: Multiple variables carry overlapping information (e.g. common driver).

Surp
: Independence surprise of joint symbol tuples.

Syn
: Residual $[\mathrm{TC} - I_{\mathrm{pair}}]_+$.

Synergy (informal)
: Variables together carry information not available from subsystems separately; excess^3 operationalises a surplus proxy for this idea.

Systemic Tau (τ_s)
: Multivariate rank-concordance thermometer of relational coherence [6].

Total correlation (TC)
: Multiinformation: total shared information among N variables [13, 4].

Window w
: Local segment length on which entropies and excess^3 are estimated.

22 One-page cheat sheet

excess^3 one-page cheat sheet (print this page)

1. **Question:** joint organisation beyond pairs in this window?
2. **Score:** excess3 $= 0.6\,\mathrm{Syn} + 0.4\,\mathrm{Surp}$.
3. **Syn:** multiinformation $-$ pairwise baseline (floor at 0).
4. **Surp:** positive surprise of joints vs independence.
5. **Primary:** continuous excess^3 and Δ.
6. **Secondary:** Φ_3 ticks (never alone at low θ_3).
7. **Weights:** fixed a priori; do not CV on the claim dataset.
8. **Null:** phase-shuffle (or cross-dependence break) on full $|\Delta|$, not mean-vs-mean alone.
9. **PID:** proxy; natural extension = synergy atom if allowed.
10. **Nesting:** parallel levels; no hard $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$.
11. **Sign:** negative Δ can mean pairwise absorption.
12. **Language:** dependence / surplus / reorganisation—not causation / "emergence proved."
13. **Encoding:** report m, τ, w, N, T (and stride if thinned).
14. **Cite for implementation:** methods paper [5]; this guide is pedagogy only.

Acknowledgements

Errors of exposition remain the author's.

Companion documents

- **Shared repository:** github.com/johelpadilla/excess3 (all three documents, figures, and validation outputs).
- **Archival DOI:** 10.5281/zenodo.21385937.
- **Methods (canonical):** *excess^3: A Pre-Specified Continuous Proxy for Order-3 Synergistic Surplus* [5].
- **Spanish introduction:** *excess^3: dependencia de orden 3 y surplus sinérgico* [7].
- **Framework:** *Systemic Tau and the RECD Framework* [6].
- **Software:** `systemictau` archival deposit [8] (`https://doi.org/10.5281/zenodo.20576241`).

Funding and competing interests

No dedicated external grant funded this guide. The author declares no competing interests.

License

References

[1] Christoph Bandt and Bernd Pompe. Permutation entropy: a natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.

[2] Nils Bertschinger, Johannes Rauh, Eckehard Olbrich, Jürgen Jost, and Nihat Ay. Quantifying unique information. *Entropy*, 16(4):2161–2183, 2014. doi: 10.3390/e16042161.

[3] Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Sonia Kéfi, Valerie Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLOS ONE*, 7(7):e41010, 2012. doi: 10.1371/journal.pone.0041010.

[4] William J. McGill. Multivariate information transmission. *Psychometrika*, 19(2):97–116, 1954. doi: 10.1007/BF02289159.

[5] Johel Padilla-Villanueva. excess^3: A pre-specified continuous proxy for order-3 synergistic surplus in the systemic tau / RECD paradigm — methods, null models, and synthetic validation, 2026. URL https://doi.org/10.5281/zenodo.21385937. Canonical methods preprint; DOI 10.5281/zenodo.21385937; repo https://github.com/johelpadilla/excess3.

[6] Johel Padilla-Villanueva. Systemic tau and the RECD framework: A relational theory of hierarchical ordinal conjunctions and critical transitions in complex systems, 2026. Companion framework manuscript; defines nested Φ_1–Φ_3 and sketches excess3.

[7] Johel Padilla-Villanueva. excess^3: dependencia de orden 3 y surplus sinérgico en sistemas multivariados, 2026. Dense Spanish introduction with integrated synthetic figures.

[8] Johel Padilla-Villanueva. systemictau: software for systemic tau and RECD analysis. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20576241. Open-source implementation; archival deposit.

[9] Fernando E. Rosas, Pedro A. M. Mediano, Michael Gastpar, and Henrik J. Jensen. Quantifying high-order interdependencies via multivariate extensions of the mutual information. *Physical Review E*, 100(3):032305, 2019. doi: 10.1103/PhysRevE.100.032305.

[10] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009. doi: 10.1038/nature08227.

[11] Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85(2):461–464, 2000. doi: 10.1103/PhysRevLett.85.461.

[12] Nicholas M. Timme and Christopher Lapish. A tutorial for information theory in neuroscience. *eNeuro*, 5(3):ENEURO.0052–18.2018, 2018. doi: 10.1523/ENEURO.0052-18.2018.

[13] Satosi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960. doi: 10.1147/rd.41.0066.

[14] Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515*, 2010.